GlycoTwin

Personalized Post-Meal Glucose Digital Twin

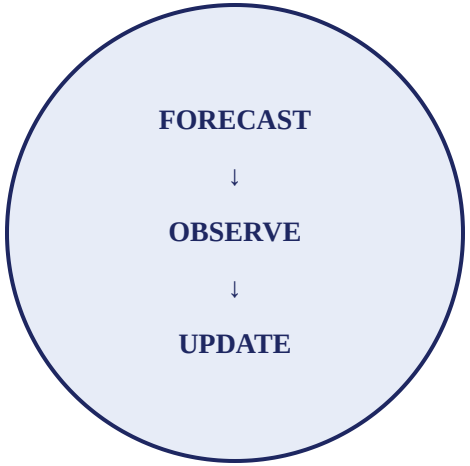
Digital Twin Challenge 2026

A stateful, uncertainty-aware Digital Twin that forecasts post-meal glucose risk, learns from each person's own eligible outcomes, and reports its evidence honestly.

Team: *[Team / member names — to be filled in]*

Institution: *[College / university — to be filled in]*

Incubator: *[Incubator name, or "Not applicable" — to be filled in]*



Problem Statement

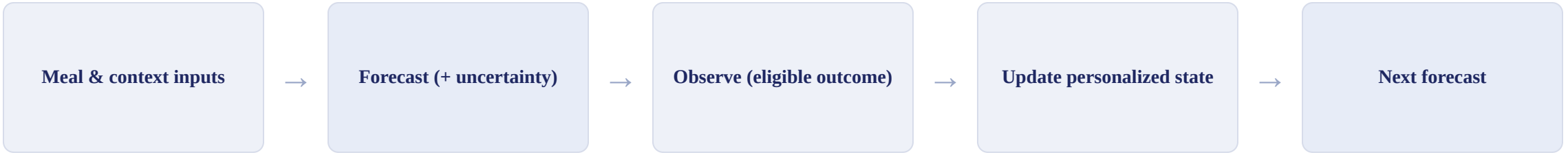
- Post-meal glucose responses vary from person to person — a single population-average forecast misses this.
- A one-time prediction does not automatically adapt when new, person-specific observations become available.
- Personalization and honest communication of uncertainty are both important research considerations, not optional extras.

Central research question

Can a stateful, personalized Digital Twin connect forecasts with eligible observations, and use them to update itself for the next forecast?

Proposed Solution

GlycoTwin keeps a persistent, versioned Digital Twin for each person and runs it through one workflow.



The twin's state carries forward — the next forecast starts from the updated parameters, not from scratch.

The model predicts the **maximum** glucose reached within the 120-minute post-meal window — not necessarily a complete glucose-over-time curve.

Innovative Features

1. **Forecast–Observe–Update**
   Connects predictions with eligible observed outcomes and subsequent state updates.

2. **Uncertainty-aware forecasting**
   Communicates prediction uncertainty — never presents an estimate as a guarantee.

3. **Bayesian-style personalization**
   Model B updates a personalized carbohydrate-response parameter from eligible observations.

4. **Persistent Digital Twin state**
   Retains model state and full version history for every subsequent forecast.

5. **Explainability & model comparison**
   Lets researchers inspect outputs and compare modelling approaches side by side.

The project's contribution is the integrated, working loop — not any single technique in isolation.

Dataset and Research Methodology

CGMacros dataset (45 participants audited) — figures below refer to different stages of the same pipeline:

1,706
logged meal rows

→

1,697
extraction-valid 120-min windows

→

1,262
core-eligible events (leakage-safe modelling set)

→

963 / 34
activity-eligible events / participants (Model B vs C)

- Meal event anchored at the logged meal row (t0); outcome window is (t0, t0+120 min].
- Features use only information available before the meal; activity measured strictly before t0.
- Leakage safeguards: participant-level folds, sequential (online) evaluation — no participant or future value leaks into a forecast.
- Not every model used every event — the 963/34 figure is the activity-eligible subset used for the Model B vs C comparison.

No participant-level health data or the dataset itself is included in this repository — aggregate, de-identified figures only.

System Architecture and Technical Stack

Frontend

React 19 · TypeScript
Vite — 7 pages, calls /api

→

Backend API

FastAPI, 127.0.0.1 only
Session + ownership on every twin route · scrypt hashing

→

Database

SQLite — 11 tables (users, twins, forecasts, observations, reconciliations…)

Research engine (shared): Bayesian Models B & C — closed-form update, forecast + uncertainty. The live twin and offline research experiments call the exact same functions.

Offline CGMacros research pipeline runs on the researcher's own machine — never reached by the website; only aggregate priors and reports cross over.

Technical stack (from the repository configuration):

- pandas, NumPy, SciPy, scikit-learn, XGBoost (Model A) — src/glycotwin/models/
- FastAPI, Pydantic v2, SQLAlchemy 2, Uvicorn, SQLite — src/glycotwin/backend/
- pytest, httpx for testing (815 Python tests recorded)

ML Models and Personalization

Model A

Population baseline (XGBoost). Trained once on the population; no personal history used.

Model B

Personalized Bayesian carbohydrate-sensitivity model. Its parameter β updates after each eligible observed meal.

Model C

Model B plus an activity-conditioning term γ — evaluated as a research comparison against Model B.

Model B: $\text{rise} = \beta \cdot \text{carbs} + \text{noise}$

Model C: $\text{rise} = (\beta + \gamma \cdot \text{activity}) \cdot \text{carbs} + \text{noise}$

- In plain English: β ("beta") is how much this person's glucose typically rises per gram of carbohydrate; γ ("gamma") adjusts that for recent activity.
- After an eligible outcome, a closed-form Bayesian update moves β toward what that meal suggests and narrows its uncertainty.
- This is a parameter update, not retraining a neural network — no model is retrained from scratch after each meal.

Research Results and Evaluation

Model B vs Model C — 963 meals, 34 participants, Libre channel, out-of-fold + sequential evaluation:

Metric	Model B	Model C	C − B	95% interval
ROC-AUC	0.7943	0.8059	+0.0116	[−0.0084, +0.0479]
Brier score	0.1391	0.1379	−0.0012	[−0.0064, +0.0030]
Rise MAE (mg/dL)	29.40	29.51	+0.11	[−0.52, +0.68]

8 of 9 paired intervals include zero → the comparison is INCONCLUSIVE

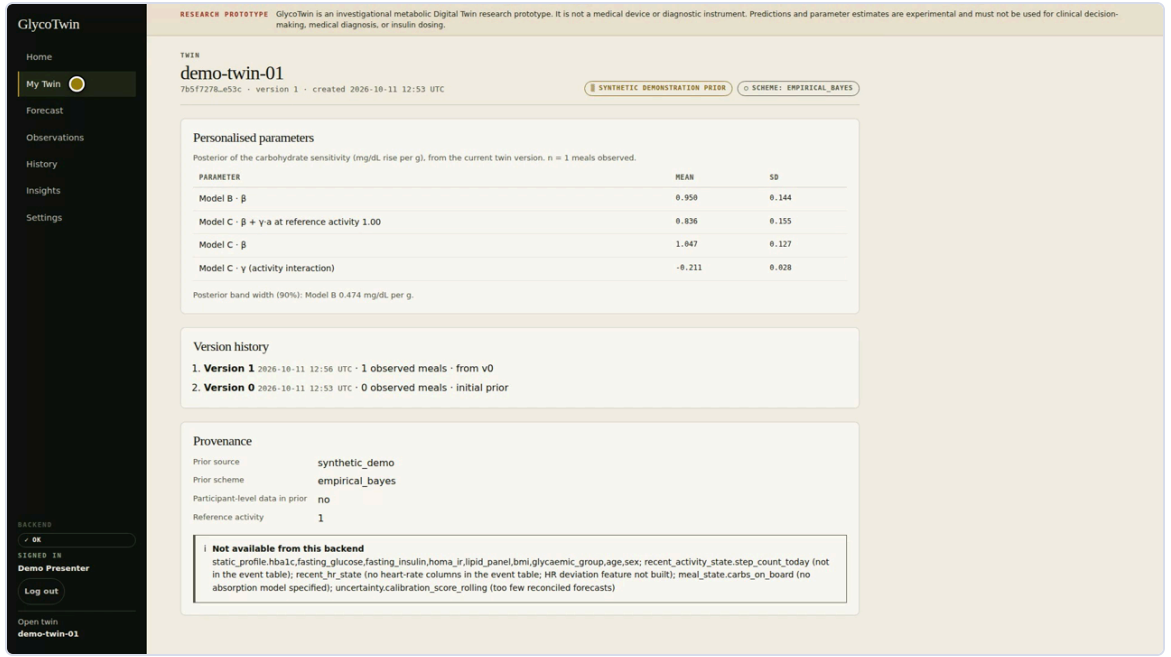
Model B does not demonstrably outperform every baseline; Model C's activity term has not been shown to help.

Three different kinds of evidence: this retrospective CGMacros result is research evidence — not the same as the live application working (shown next slide), and neither is the same as clinical validation, which has not been done.

Demonstration, Limitations and Safety

- Forecast (version 0): P(max ≥ 180 mg/dL) = 7.7%
- Observe: an eligible outcome is recorded after the 120-min window closes
- Update: Model B β moves from 0.747 → 0.950 (new twin version 1)
- Next forecast, same inputs: P(max ≥ 180 mg/dL) = 21.0%

Same inputs, different twin state, different forecast.



Real application screenshot — My Twin page after one eligible update (version 0 → 1)

Limitations and safety

- Investigational research prototype — not a clinically validated diagnostic or treatment system.
- Not intended for insulin dosing or any clinical decision.
- Further evaluation and independent validation are required before any real-world use.

Conclusion and Future Work

- Problem: one-off glucose predictions cannot adapt to the individual.
- Central innovation: a persistent, versioned Digital Twin that links forecasting, uncertainty, and sequential Bayesian personalization in one working loop.
- The retrospective CGMacros research gives early, honest evidence — inconclusive for the activity-conditioning hypothesis, but the pipeline and safeguards are validated.
- Next steps: run the full prespecified key experiment on real data with baseline controls; test whether an activity effect depends on activity level.

Repository: github.com/medhachoudhari/GlycoTwin

Team / Institution: *[to be filled in — see title slide]*

GlycoTwin connects forecasting, observation, and personalization into one honest, auditable research workflow.

Investigational research prototype — not a medical device, not clinically validated.